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Saveetha Institute of Medical And Technical Sciences

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ITA0329-MOBILE COMPUTING

"Mobile IP Implementation with Home/Foreign Agent Simulation"

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INDEX

- ABSTRACT
- INTRODUCTION
- PROBLEM STATEMENT AND PROPOSED SOLUTION
- MODULES
- RESULTS
- CONCLUSION
- REFERENCES

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ABSTRACT

- Mobile IP allows devices to move between networks while keeping their permanent home IP address.
- The simulation includes Mobile Node, Home Agent, Foreign Agent, and Correspondent Node.
- The Mobile Node registers its Care-of Address with the Home Agent.
- The Home Agent tunnels packets to the Mobile Node through the foreign network.
- The Foreign Agent assists packet delivery.
- The system maintains continuous communication during network movement.

INTRODUCTION

- Mobile IP supports device mobility across different IP networks.
- The Mobile Node retains its permanent home IP address.
- The Home Agent tracks the device’s location, while the Foreign Agent assists packet delivery.
- The simulation demonstrates registration and Care-of Address mapping.
- Packet tunneling enables successful packet delivery.
- The system maintains continuous communication during network movement.

PROBLEM STATEMENTS AND PROPOSED SOLUTION

Problem Statement:

- Network changes may interrupt communication.
- The device's current location may be unknown.
- Packets may fail to reach the Mobile Node.

Proposed Solution:

- Simulate movement to a foreign network.
- Register the Care-of Address with the Home Agent.
- Tunnel packets through the Foreign Agent for successful delivery.

Module 1: Mobile Node and Agent Registration

Purpose: Register the Mobile Node's location in a foreign network.

- The Mobile Node retains its home IP address.
- It discovers the Foreign Agent and obtains a Care-of Address.
- The Home Agent registers and stores the address mapping.

Input: Mobile Node's home IP address, Foreign Agent details, and Care-of Address. **Output:** successful registration and an updated mobility binding.

Module 2: Tunneling and Packet Delivery

Purpose: The Correspondent Node sends packets to the Mobile Node's home IP.

- The Home Agent encapsulates and tunnels packets to the Care-of Address.
- The Foreign Agent forwards packets to the Mobile Node.
- The simulation displays packet details and delivery status.

Input: Packet to the home IP.

Output: packets reach the Mobile Node without changing its permanent home address.

RESULTS

- The Mobile IP simulation successfully demonstrated Mobile Node registration and communication across different IP networks.
- The Home Agent stored the Care-of Address mapping and tracked the Mobile Node’s current location.
- Packet tunneling enabled successful delivery through the Foreign Agent to the Mobile Node.
- Overall, the system maintained the permanent home IP address and supported continuous communication during network movement.

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CONCLUSION

- Mobile IP enables devices to move between different IP networks while maintaining their permanent home IP address.
- The Home Agent tracks the Mobile Node’s location using Care-of Address registration.
- Packet tunneling ensures successful packet delivery through the Foreign Agent.
- Overall, the simulation demonstrates continuous communication during network mobility.

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Thank you!